Je'aime H. Powell

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Professional Summary

A seasoned IT professional with over 25 years of experience in High-Performance Computing (HPC), Cyberinfrastructure, and Workforce Development. Proven track record in leading large-scale systems administration, spearheading research initiatives, and mentoring cross-functional teams. Adept at designing and implementing cloud-based infrastructures, with a strong commitment to innovation and excellence. Seeking to leverage deep technical expertise and leadership experience in a senior managerial role.

Core Competencies

- **High-Performance Computing Administration(HPC)**
- **Cloud Infrastructure Management**
- **Cybersecurity**
- **Research Design and Implementation**
- **Workforce Development**
- **Project Management**
- **Cyberinfrastructure Research**
- **IoT Sensor Research**
- **Hackathon Professional**

Professional Experience

Omnibond Systems
Orlando, Florida
Product Analyst and Outreach Coordinator
October 2024 – present

- Used data to understand customer perception of products, identified usability issues, and suggested improvements.
- Gathered and analyzed data from various sources including user feedback, market research, and product usage metrics to inform product strategy.
- Analyzed competitor products and strategies to identify opportunities for differentiation.
- Worked closely with cross-functional teams like engineering, marketing, sales, and customer support to ensure successful product delivery and adoption.
- Provided technical input into multiple aspects of products, including software, hardware, and web-based products.
- Maintained a calendar of outreach activities, including community events, workshops, and appearances.
- Maintained accurate records and electronic files for education and outreach efforts.
- Continued to develop active partnerships with community leaders and service groups.

Texas Advanced Computing Center (TACC)
Austin, Texas
Senior Systems Administrator
April 2017 – September 2024

- Managed and expanded production storage resources, including Cray supercomputing hardware.
- Supported Linux clusters, diagnosing performance issues, and resolving system emergencies.
- Developed comprehensive documentation aligned with TACC's workflow tracking methodology.
- Mentored junior staff and led cross-functional teams in system administration.
- Led research initiatives on Environmental IoT Sensors and Decision Support Services.
- Designed and supported cloud-based infrastructure using AWS, Google Cloud, and Azure.
- Developed and deployed low-cost environmental sensors for various research projects.
- Chaired the HackHPC organizing committee, fostering interdisciplinary training and workforce development.

The University of Texas at Austin
Austin, Texas
University Lecturer
January 2018 – Present
- Taught courses on Scientific Programming, emphasizing C and Fortran languages for scientific applications.

Helios LLC

Denver, Colorado
IoT Sensors Specialist
October 2022 – June 2023
- Designed and tested IoT sensors for enhanced product functionality and data analysis. - Developed smart contracts on the Ethereum blockchain, ensuring data security and integrity.

Cray Inc.
Orlando, Florida
Systems Analyst
April 2015 – April 2017
- Managed a secured Cray XC40 supercomputer, ensuring optimal performance and security. - Developed technical documentation and led system maintenance initiatives.

Old Dominion University
Norfolk, Virginia
High Performance Computing Systems Administrator
February 2014 – April 2015
- Managed HPC facilities, including user management and storage optimization. - Delivered HPC training programs and spearheaded the implementation of the Globus Connect Server.

Elizabeth City State University
Elizabeth City, North Carolina
Grid Manager
July 2006 – January 2014
- Administered Linux Computing Cluster, leading initiatives in system management and research mentorship.

Education & Certifications

Elizabeth City State University
Master's Degree, Applied Mathematics with a Concentration in Remote Sensing
2007 – 2010

Elizabeth City State University
Bachelor's Degree, Computer Science
1997 – 2004

Certifications

- A+ Certification

- Network+ Certification
- Security+ Certification
- Cray XC30 Administrator and Hardware Support Certification
- Amateur Radio Operator (KG5TQW)

Awards & Recognitions

- **Co-Principal Investigator** - [Cybertraining Pilot] HPC-ED: Building a Federated Repository and Increasing Access through Cybertraining (NSF Award: 2320977)
- **Principal Investigator** - μP-STREAM: Micro-controller Platform Sending Telemetry Real-time for Earth's Adaptive Models
- **Principal Investigator** - Cloudify Gateways: Federated Science Gateway for On-Demand Research Analysis and Collaborative Knowledge (ODSACK)
- **Science Gateways Community Institute's Young Professional of the Year Awardee**
- **BEYA Graduate Leadership Award**
- **PolarGrid Greenland Field Deployment Team**
- **Canon Information Technology Services Presidential Excellence Award**

Technical Skills

- **Programming Languages:** C, Fortran, Python, C++, Shell Scripting, HTML5/CSS, JavaScript, Jekyll
- **Operating Systems:** Linux (Red Hat, SLES, CentOS, Ubuntu), Windows, macOS
- **HPC System Software:** Cray Linux Environment, xCAT, Condor, Bright Cluster Manager, Ansible, Jenkins, SLURM, PBS/Torque, SGE, LSF, GPFS, Lustre, Globus, IPMI, LoM, LMoD, Singularity
- **Cloud Platforms:** Amazon Web Services(AWS), Google Cloud Platform (GCP), Microsoft Azure, Omnibond's Project Eureka
- **Security:** Cybersecurity protocols
- **Tools:** Git/GitHub, Docker, Kubernetes, Jupyter Hub/Notebooks
- **Networking:** TCP/IP(VLAN), Ethernet, InfiniBand, Fiber Channel
- **Database Management:** PostgreSQL, MySQL
- **IoT:** Arduino, Raspberry Pi, Sensor Networks, MQTT, LoRA, Particle, Backend intergration, Environmental monitoring, Sensor design
- **Other:** Strategic Planning, Technical Documentation, Mentoring, Technical Training, Workforce Development, Education and Outreach

Publications and Presentations

1. M. P. Thomas et al., “Building a Federated Catalog for CyberTraining Materials: The HPC-ED Pilot Project,” in Practice and Experience in Advanced Research Computing 2024: Human Powered Computing, in PEARC ’24. New York, NY, USA: ssociation for Computing Machinery, 2024. doi: 10.1145/3626203.3670586.
2. S. Mehringer et al., “Scaling HPC Education,” The Journal of Computational Science Education, vol. 15, no. 1, pp. 41–46, Mar. 2024, doi: https://doi.org/10.22369/issn.2153-4136/15/1/8.
3. J. K. Holmen et al., “FacultyHack Events: Faculty-Focused Hackathons for High-Performance Computing Curriculum Development,” in Proceedings of the 8th International Conference on Game Jams, Hackathons and Game Creation Events, openhagen Denmark: ACM, Oct. 2024, pp. 67–71. doi: 10.1145/3697789.3697796.
4. A. Contina et al., “BioSense: An automated sensing node for organismal and environmental biology,” HardwareX, vol. 20, p. e00584, Dec. 2024, doi: 10.1016/j.ohx.2024.e00584.
5. S. Mehringer et al., “Assessing Shared Material Usage in the High Performance Computing (HPC) Education and Training Community,” The Journal of Computational Science Education, vol. 14, no. 2, pp. 18–22, Nov. 2023, doi: 10.22369/ssn.2153-4136/14/2/4.
6. J. Powell, C. Dey, J. Powell, and D. Sarojini, “[Workshop. Pandemic! Control a Pandemic Disease before it Runs Rampant,” in Supercomputing 22, 2022.
7. A. O. Medina and J. Powell, “Methodology for Imagery Metadata Collection and Entry into a PostgreSQL Database using Stampede2,” in Practice and Experience in Advanced Research Computing, ACM, Jul. 2022. doi: 10.1145/3491418.535137.
8. Jeaime Powell, L. B. Hayden, A. Cannon, J. Holly, C. Dey, and A. Nolte, “The HackHPC Model: Fostering Workforce Development in High-Performance Computing Through Hackathons,” 2022, doi: 10.5281/ZENODO.7089496.
9. J. Powell, L. Bailey Hayden, A. Cannon, B. Wilson, and A. Nolte, “Organizing Online Hackathons for Newcomers to a Scientific Community – Lessons Learned from Two Events,” in Sixth Annual International Conference on Game Jams, ackathons, and Game Creation Events, in ICGJ 2021. New York, NY, USA: Association for Computing Machinery, 2021, pp. 78–82. doi: 10.1145/3472688.3472700.
10. Y. Gil et al., “Towards Democratizing Modeling at Scale,” in 2021 IEEE 17th International Conference on eScience (eScience), 2021, pp. 269–270. doi: 10.1109/eScience51609.2021.00054.
11. S. C. Dey et al., “Transitioning Education and Training to a Virtual World, Lessons Learned,” The Journal of Computational Science Education, vol. 12, no. 2, pp. 18–20, Feb. 2021, doi: 10.22369/issn.2153-4136/12/2/3.
12. J. Powell, J. Powell, P. Marchetto, A. M. Matheny, and S. A. Pierce, “The Conversion of a Multipoint Data Logging System into a Mesh-Based Observation Network: Implementation

of a Mesh-Based Observation Network," in MOXXI 2019 nternational Conference "Innovation in Hydrometry: Overcoming Barriers to Operationalization," 2019.
13. J. Powell, S. Cleveland, J. Powell, J. Stubbs, S. A. Pierce, and M. Daniels, "Streamed Data via Cloud-Hosted Real-Time Data Services for the Geosciences as an Ingestion Interface into the Planet Texas Science Gateway and ntegrated Modeling Platform," in Science Gateways 2019, 2019.
14. A. M. Matheny et al., "LEAF: Logger for ecological and atmospheric factors," HardwareX, vol. 6, p. e00079, Oct. 2019, doi: 10.1016/j.ohx.2019.e00079.
15. S. B. Cleveland et al., "Tapis-CHORDS Integration: Time-Series Data Support in Science Gateway Infrastructure," in Conference: Science Gateways 2019At: San Diego, CA, 2019.
16. M. J. Powell, L. Hayden, C. Farmer, and K. McDonald, "TeraScan Curriculum Development and Integration of SeaSpace Technology into the Classroom," in Earth Observation: Education, Jul. 2014. [Online]. Available: https://www2.ecurecms.com/IGARSS2014/Papers/viewpapers.asp?papernum=1216
17. I. K. Williams, R. A. Armaah, K. Hayden, and others, "Development of an algorithm to predict coastal buoy temperature from Advanced Very High Resolution Radiometer (AVHRR)," in 2012 IEEE International Geoscience and Remote ensing Symposium, IEEE, 2012, pp. 2625–2628.
18. L. Hayden, B. Aldrich Jr, M. Je'aime Powell, and C. Huff, "Spectroscopic Image Signature Classification of Land Cover Types using Multi-Spectral Data within a Neural Network," 2012.
19. N. Dykes, L. Easley, J. E. Mitchell, K. L. Hayden, and others, "Charles creek flood zone modeling: A correlation study of environmental conditions versus water level in the Pasquotank watershed," in 2012 IEEE International eoscience and Remote Sensing Symposium, IEEE, 2012, pp. 966–969.
20. M. Austin, N. Barmore, A. Brumfield, J. Deloatch, J. Powell, and L. H. PI, "Integration With A Web Application To Create Navigational Instructions For Locations On The Campus Of Elizabeth City State University," 2012.
21. J. Powell and L. Hayden, "Multi-Channel RADAR Depth Sounder (MCRDS) signal processing: A distributed computing approach," in 2010 IEEE International Geoscience and Remote Sensing Symposium, Honolulu, HI, USA: IEEE, Jul. 2010, pp. 358–3361. doi: 10.1109/IGARSS.2010.5651272.
22. L. Hayden, A. Jearld, K. Hayden, N. L. Jackson, and others, "Hands-on GPS and remote sensing training for high school learners during IGARSS 2009 in Cape Town, South Africa," in 2010 IEEE International Geoscience and Remote ensing Symposium, IEEE, 2010, pp. 1118–1121.
23. L. Hayden, J. H. Powell, and E. Akers, "Establishing field and base camp servers for remote sensing of ice sheets in ilulissat, Greenland," in 2009 IGARSS, IEEE, 2009. doi: 10.1109/igarss.2009.5418148.